



Event Management Mobile Application for Attendee Self-Service

Team 23 Section 12

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Abstract

- A web platform for attendees to discover events, book seats and access QR tickets.
- React, Express and MySQL support digital registration and organizer management.

Problem Statement & Objectives

- Problem: Manual registration, paper tickets and long entry queues.
- Objective: Enable online event booking.
- Objective: Verify entry with QR tickets.
- Objective: Centralize schedules and updates.
- Scope: Attendees and event administrators.

Existing vs Proposed System

Feature	Existing System	Proposed System
Sign-up	Manual forms	Online account
Ticketing	Paper passes	Digital QR ticket
Entry	Manual checking	QR validation
Updates	Scattered details	Schedules and alerts
Seat control	Manual counting	Database row locks

Feasibility Study

- Technical: React, Express and MySQL support a browser-based application.
- Economic: Open-source development tools support a low-cost prototype.
- Operational: Self-service reduces repeated organizer assistance.
- Schedule: Develop and review the core modules in stages within the semester.

Results & Validation

- Digital booking and QR entry flow.
- Review checks: login, event CRUD, seat capacity and duplicate check-in.

Conclusion

- One platform connects event discovery, booking and entry.
- Self-service reduces paperwork and manual coordination.

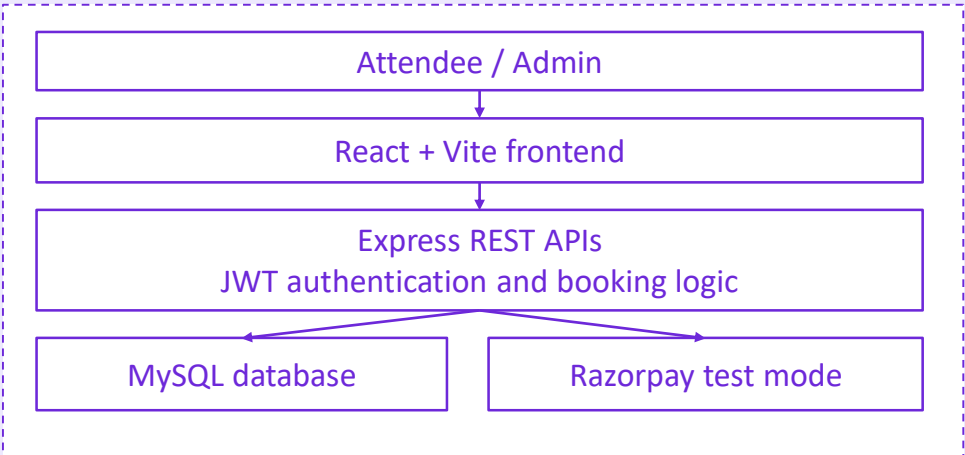
Future Scope

- Event recommendations based on interests.
- Maps, push reminders and richer attendance analytics.

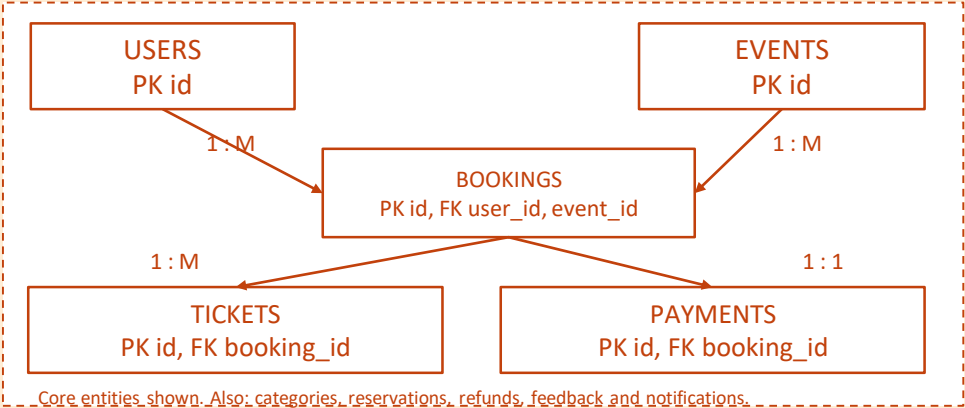
Tools & Technologies

- Front-End: React 18 and Vite
- Back-End: Node.js, Express and REST APIs
- Database: MySQL 8.0 / MariaDB
- Security: JWT and bcrypt password hashing
- Payments: Razorpay test-mode checkout
- Ticketing: QR code generation and scanning

System Architecture



ER Diagram & Database Design



Implementation Features

ATTENDEE	ADMIN
Register and log in	Create and edit events
Search events	Manage attendees
Book and pay	Scan QR for check-in
Download QR ticket	Reject duplicate entry
Cancel and track refunds	Send announcements